

# CH:6 NON-PARAMETRIC TECHNIQUES FOR DENSITY ESTIMATION

## Contents

|                                                    |   |
|----------------------------------------------------|---|
| Density Estimation.....                            | 1 |
| Parametric Methods of Density Estimation.....      | 1 |
| Non-Parametric Techniques .....                    | 2 |
| Key Terminologies.....                             | 3 |
| Types of Distances .....                           | 4 |
| K-Nearest Neighbours (KNN) method.....             | 5 |
| Parzen Window/Kernel-density Estimate Method ..... | 6 |

## Density Estimation

Density estimation is the process of **estimating the probability distribution** for a dataset. In other words, we try to find how data is distributed across the space without assuming any fixed shape or formula for the distribution.

### Parametric Methods of Density Estimation

Parametric methods assume that the data follows a certain fixed and known probability distribution (like normal, exponential, etc.) characterized by a set of parameters. The goal is to estimate these parameters from the data.

- The form of the probability distribution is **known in advance**.
- The entire density estimation problem reduces to **estimating a finite set of parameters** (e.g., mean, variance).
- These methods are **efficient** when the assumption about the underlying distribution is correct.
- The complexity of the model is **independent of the amount of data**.
- These are often easier to **interpret** and **compute**.

#### Process Overview:

1. Assume a specific functional form for the probability density function.
2. Use a statistical method (like maximum likelihood) to estimate the parameters of that function based on observed data.
3. Once parameters are estimated, the full distribution is defined.

#### Advantages:

- Requires less data for good performance (if assumptions hold).
- Simpler and computationally faster.
- Easier mathematical analysis.

#### Disadvantages:

- Performance heavily depends on the **correctness of the assumed distribution**.
- **Less flexible** when the actual data distribution deviates from the assumed form.

### Examples Parametric Methods:

- MLE
- EM algorithm
- Markov models
- Gaussian Distribution
- Exponential Distribution
- Poisson Distribution
- Bernoulli Distribution
- Binomial Distribution

### Non-Parametric Techniques

Non-parametric density estimation methods do **not assume a predefined functional form** (like Gaussian, Poisson, etc.) for the underlying distribution. Instead, they build the distribution **directly from the data**, adapting to the actual structure of the dataset.

- **No fixed number of parameters:** Unlike parametric models, these methods can adjust complexity based on the data.
- **Data-driven:** The model's shape is determined by the data points themselves.
- **Flexible:** Can model complex and irregular distributions that don't fit standard parametric forms.
- **Local estimation:** Often, the density is estimated in small regions around each point.

### Process Overview:

1. Use the data to construct an approximation of the underlying probability distribution.
2. The method relies on **local properties of data** (e.g., frequency counts in regions or data proximity).
3. The estimated density improves as the **amount of data increases**.

### Advantages

1. **No assumption:** Doesn't assume any fixed distribution form.
2. **Flexible modeling:** Adapts to any data shape, including complex ones.
3. **Data-driven:** Entirely based on actual data points.
4. **Works for unknown distributions:** Effective when distribution is not known.
5. **Good for visualizing data:** Useful in identifying clusters or patterns.

### Disadvantages:

1. **Slow for large data:** Computational cost is high with big datasets.
2. **Parameter sensitivity:** Results depend on window size/k/bandwidth.
3. **Not scalable:** Struggles with high-dimensional data.
4. **Less interpretable:** No simple parameters to describe the model.
5. **Risk of overfitting:** May fit noise if not tuned properly.

### Non-Parametric Methods:

- KNN
- Parzen Window Estimation
- Histogram
- Kernel Density Estimation (KDE)
- Empirical Distribution Function (EDF)

| Feature                     | Parametric Methods                    | Non-Parametric Methods                   |
|-----------------------------|---------------------------------------|------------------------------------------|
| Model Assumption            | Assumes a fixed distribution form     | No fixed form assumed                    |
| Examples                    | Gaussian, Exponential distributions   | Histogram, Kernel Density Estimation     |
| Parameters                  | Finite (e.g., mean, variance)         | Infinite or flexible parameters          |
| Flexibility                 | Less flexible                         | More flexible                            |
| Accuracy (for unknown data) | May be poor if assumption is wrong    | Can adapt better to data shape           |
| Computation                 | Often simpler                         | More complex, especially with large data |
| Data Requirement            | Requires fewer data points            | Requires more data for accuracy          |
| Overfitting                 | Less prone                            | Can overfit if not tuned properly        |
| Interpretability            | Easier to interpret                   | Harder to interpret                      |
| Use Case                    | When distribution is known or assumed | When distribution is unknown             |

## Key Terminologies

1. **Hyperparameters** are values set before training a model that control its learning process, like the learning rate or number of neighbours in KNN.
2. **Parameter tuning** refers to the process of selecting the optimal values for the hyperparameters of a model (e.g.,  $k$  in KNN) to improve its performance and generalization on unseen data.
3. **Model**: A mathematical representation used to make predictions from data.
4. **Training**: The process of teaching a model using labelled data.
5. **Testing**: Evaluating a trained model on new, unseen data.
6. **Overfitting**: When a model learns too much from the training data, capturing noise rather than patterns.
7. **Underfitting**: When a model is too simple to capture the underlying patterns in the data.
8. **Feature**: An individual measurable property or characteristic of the data.
9. **Label**: The output or target value corresponding to a data point.
10. **Loss Function**: A function that measures how well a model's predictions match the actual data.
11. **Accuracy**: The percentage of correct predictions made by the model.
12. **Precision**: The proportion of true positive predictions out of all positive predictions.
13. **Bias**: The error introduced by approximating a real-world problem with a simplified model.
14. **Variance**: The model's sensitivity to small fluctuations in the training data.
15. **Clustering**: Grouping data points into clusters where points in the same group are more similar to each other than to those in other groups.

## Types of Distances

1. **Euclidean Distance:** The most common distance metric, used to compute the straight-line distance between two points in Euclidean space.

|                                                                 |                                                                                                                                                                                     |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$ | <p>Given two points <math>\mathbf{x} = (3, 4)</math> and <math>\mathbf{y} = (0, 0)</math>:</p> $d(\mathbf{x}, \mathbf{y}) = \sqrt{(3-0)^2 + (4-0)^2} = \sqrt{9+16} = \sqrt{25} = 5$ |
|-----------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

2. **Manhattan Distance (L1 Norm):** Computes the sum of the absolute differences between corresponding coordinates. Also known as **taxicab** or **city block distance**.

|                                                        |                                                                                                                                                |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^n  x_i - y_i $ | <p>For points <math>\mathbf{x} = (3, 4)</math> and <math>\mathbf{y} = (0, 0)</math>:</p> $d(\mathbf{x}, \mathbf{y}) =  3-0  +  4-0  = 3+4 = 7$ |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|

3. **Minkowski Distance:** A generalization of Euclidean and Manhattan distances. It's defined by a parameter  $p$  which controls the type of distance:

- ☐ When  $p=1$ , it becomes **Manhattan Distance**.
- ☐ When  $p=2$ , it becomes **Euclidean Distance**.

|                                                                                       |                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $d(\mathbf{x}, \mathbf{y}) = \left( \sum_{i=1}^n  x_i - y_i ^p \right)^{\frac{1}{p}}$ | <p>For <math>p = 3</math>, <math>\mathbf{x} = (3, 4)</math>, and <math>\mathbf{y} = (0, 0)</math>:</p> $d(\mathbf{x}, \mathbf{y}) = ( 3-0 ^3 +  4-0 ^3)^{\frac{1}{3}} = (27 + 64)^{\frac{1}{3}} = (91)^{\frac{1}{3}} \approx 4.49$ |
|---------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

4. **Cosine Similarity:** Measures the cosine of the angle between two vectors. It's often used for text data, where vectors represent word frequencies or term weights.

|                                                                                                |                                                                                                                                                                                                                                                                                                                                                                             |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\text{cosine similarity} = \frac{\mathbf{x} \cdot \mathbf{y}}{\ \mathbf{x}\  \ \mathbf{y}\ }$ | <p>For vectors <math>\mathbf{x} = (1, 2, 3)</math> and <math>\mathbf{y} = (4, 5, 6)</math>:</p> $\mathbf{x} \cdot \mathbf{y} = 1 \times 4 + 2 \times 5 + 3 \times 6 = 32$ $\ \mathbf{x}\  = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}, \quad \ \mathbf{y}\  = \sqrt{4^2 + 5^2 + 6^2} = \sqrt{77}$ $\text{cosine similarity} = \frac{32}{\sqrt{14} \times \sqrt{77}} \approx 0.974$ |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

5. **Hamming Distance:** Used primarily for categorical variables, it counts the number of positions at which the corresponding elements of two strings are different.

|                                                                                                                                                                        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>For strings <math>\mathbf{x} = "1101"</math> and <math>\mathbf{y} = "1011"</math>:</p> $\text{Hamming distance} = 2 \quad (\text{positions 2 and 3 are different})$ |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

6. **Chebyshev Distance:** Similar to Manhattan distance but takes the maximum difference along any coordinate dimension.

|                                                 |                                                                                                                                                     |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| $d(\mathbf{x}, \mathbf{y}) = \max( x_i - y_i )$ | <p>For <math>\mathbf{x} = (3, 4)</math> and <math>\mathbf{y} = (0, 5)</math>:</p> $d(\mathbf{x}, \mathbf{y}) = \max( 3-0 ,  4-5 ) = \max(3, 1) = 3$ |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|

7. **Mahalanobis Distance:** A distance metric that accounts for the correlations of the data set and is used for normalized data.

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})^T S^{-1} (\mathbf{x} - \mathbf{y})}$$

Consider  $\mathbf{x} = (2, 3)$ ,  $\mathbf{y} = (1, 1)$ , and the covariance matrix  $S = \begin{bmatrix} 2 & 0.5 \\ 0.5 & 1 \end{bmatrix}$ . First, compute the inverse of  $S$ :

$$S^{-1} = \begin{bmatrix} 1.2 & -0.6 \\ -0.6 & 2 \end{bmatrix}$$

Now calculate the Mahalanobis distance:

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{(\mathbf{x} - \mathbf{y})^T S^{-1} (\mathbf{x} - \mathbf{y})}$$

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\begin{bmatrix} 1 \\ 2 \end{bmatrix}^T \begin{bmatrix} 1.2 & -0.6 \\ -0.6 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix}}$$

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{2.4 + 4.8} = \sqrt{7.2} \approx 2.68$$

## K-Nearest Neighbours (KNN) method

**KNN** is a **supervised, non-parametric, lazy learning algorithm** used for both **classification** and **regression**.

1. **k** is a parameter that refers to the number of nearest neighbours considered when predicting the class of a new data point.
2. **Type:** Supervised learning algorithm — used for classification and regression.
3. **Nature:** Non-parametric and lazy — no model is built during training.
4. **Training:** No explicit training phase — stores the dataset.
5. **Prediction:** Based on majority vote (classification) or average (regression) of **k** nearest neighbors.
6. **Distance metric:** Typically uses Euclidean distance, but others like Manhattan or Minkowski can also be used.
7. **Feature scaling:** Necessary — unscaled features can distort distance calculations.
8. **Choice of k:** A small **k** may lead to overfitting; a large **k** can smooth the prediction but may miss local patterns.

### Advantages:

1. **Simple and easy to implement** with minimal parameters to tune.
2. **No training phase** – making it fast to deploy.
3. **Flexible:** Can be used for both classification and regression tasks.
4. **Naturally handles multi-class** classification problems.
5. **Works well with non-linear decision boundaries.**

### Disadvantages:

1. **Computationally expensive** during prediction, especially with large datasets.
2. **Sensitive to irrelevant features** and requires feature scaling.
3. **Curse of dimensionality:** Performance degrades in high-dimensional spaces.
4. **Needs large memory** since it stores the entire dataset.
5. **Sensitive to noise** in the dataset, which can affect prediction accuracy.

**Example:**

Predict BMI category for height= 165 and weight =75 when value of k =

- i) 3
- ii) 5
- iii) 7

| Height (cm) | Weight (kg) | BMI Category  |
|-------------|-------------|---------------|
| 160         | 55          | Normal weight |
| 165         | 65          | Normal weight |
| 170         | 70          | Normal weight |
| 175         | 85          | Overweight    |
| 180         | 90          | Overweight    |
| 185         | 110         | Obesity       |
| 190         | 115         | Obesity       |
| 150         | 40          | Underweight   |
| 155         | 50          | Underweight   |
| 200         | 120         | Obesity       |

**Answer**

| Height | Weight | BMI Category  | Distance Calculation                                                              | Rank |
|--------|--------|---------------|-----------------------------------------------------------------------------------|------|
| 160    | 55     | Normal weight | $\sqrt{((165-160)^2 + (75-55)^2)} = \sqrt{(25 + 400)} \approx \mathbf{20.62}$     | 4    |
| 165    | 65     | Normal weight | $\sqrt{((165-165)^2 + (75-65)^2)} = \sqrt{(0 + 100)} = \mathbf{10.00}$            | 2    |
| 170    | 70     | Normal weight | $\sqrt{((165-170)^2 + (75-70)^2)} = \sqrt{(25 + 25)} \approx \mathbf{7.07}$       | 1    |
| 175    | 85     | Overweight    | $\sqrt{((165-175)^2 + (75-85)^2)} = \sqrt{(100 + 100)} \approx \mathbf{14.14}$    | 3    |
| 180    | 90     | Overweight    | $\sqrt{((165-180)^2 + (75-90)^2)} = \sqrt{(225 + 225)} \approx \mathbf{21.21}$    | 5    |
| 185    | 110    | Obesity       | $\sqrt{((165-185)^2 + (75-110)^2)} = \sqrt{(400 + 1225)} \approx \mathbf{40.31}$  | 8    |
| 190    | 115    | Obesity       | $\sqrt{((165-190)^2 + (75-115)^2)} = \sqrt{(625 + 1600)} \approx \mathbf{47.17}$  | 9    |
| 150    | 40     | Underweight   | $\sqrt{((165-150)^2 + (75-40)^2)} = \sqrt{(225 + 1225)} \approx \mathbf{38.08}$   | 7    |
| 155    | 50     | Underweight   | $\sqrt{((165-155)^2 + (75-50)^2)} = \sqrt{(100 + 625)} \approx \mathbf{26.93}$    | 6    |
| 200    | 120    | Obesity       | $\sqrt{((165-200)^2 + (75-120)^2)} = \sqrt{(1225 + 2025)} \approx \mathbf{57.01}$ | 10   |
| 165    | 75     |               |                                                                                   |      |

| k | Top k Persons           | Predicted BMI Category |
|---|-------------------------|------------------------|
| 3 | N, N, OW                | Normal weight          |
| 5 | N, N, OW, N, OW         | Normal weight          |
| 7 | N, N, OW, N, OW, UW, UW | Normal weight          |

## Parzen Window/Kernel-density Estimate Method

The Parzen Window method is a non-parametric density estimation technique used to estimate the probability density function (PDF) of a random variable. It works by placing a smooth kernel (window function) on each data point and averaging them to get the overall density.

- **Non-parametric:** No assumption about underlying distribution.
- **Smooth:** Produces a continuous and differentiable density estimate.
- **Local influence:** Nearby points contribute more to the estimate at xxx.
- **Kernel Function K:** A smooth function that assigns weights to data points based on their distance from the point of estimation like: Uniform, Gaussian
- **Bandwidth h:** a parameter that controls the width of the window, affecting smoothness of the density estimate
  - **Small h:** high variance, low bias (more sensitive)
  - **Large h:** low variance, high bias (over-smoothing)

**How it works?**

- A kernel (usually symmetric like a Gaussian or uniform box) is centered at each data point.

- The kernel's height and spread are determined by the bandwidth  $h$ .
- The value of the density at any point  $x$  is the sum of all the kernel values at that point.
- As a result, the estimated density is smoother as  $h$  increases.

### Basic Idea:

- Suppose you have a dataset with  $n$  data points:  $x_1, x_2, \dots, x_n$
- You want to estimate the **density function**  $p(x)$  at a point  $x$ .

The estimate is given by:

$$\hat{p}(x) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

Where:

- $K$ : the **kernel function** (e.g., Gaussian, rectangular)
- $h$ : the **window width** or **bandwidth** (smoothing parameter)
- $d$ : the number of dimensions of the data
- $x_i$ : the sample data points

### Advantages:

1. **Non-parametric**: Doesn't assume any fixed distribution form.
2. **Flexible**: Can model complex and multimodal distributions.
3. **Smooth estimation**: Produces continuous and differentiable density estimates.
4. **Intuitive**: Easy to understand and implement.
5. **Adaptable**: Works well for small and unknown datasets.

### Disadvantages of Parzen Window Method:

1. **Bandwidth sensitivity**: Performance heavily depends on bandwidth choice.
2. **Computationally expensive**: High computational cost for large datasets.
3. **Curse of dimensionality**: Struggles in high-dimensional spaces.
4. **Choice of kernel matters**: Kernel selection can affect results.
5. **Over-smoothing**: Large bandwidth can lead to over-smoothing and loss of detail.

### Derivation of Gaussian Kernel density Estimator (KDE)

General KDE  $\Rightarrow p(x) = \frac{1}{nh^d} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$  — (i)

Gaussian kernel:  $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$  — (ii)

Substituting (ii) in (i)

$$K(u) = K\left(\frac{x - x_i}{h}\right)$$

$$\Rightarrow u = \frac{x - x_i}{h}$$

$$\therefore p(x) = \frac{1}{nh^d \sqrt{2\pi}} \sum_{i=1}^n e^{-\frac{(x - x_i)^2}{2h^2}}$$

## ◆ Question

You are given a 1D dataset of sample points:  $x = [1.0, 1.5, 2.0, 3.0, 3.5]$

Use the Parzen Window density estimation method with the following settings:

- Window width ( $h$ ) = 1.0
- Kernel: Use the standard normal (Gaussian) kernel  $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$

Tasks: Estimate the probability density at point  $x = 2.5$ .

Answer

$n = 5$      $h = 1$      $K(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$   
 $= 0.3989 e^{-\frac{u^2}{2}}$

①  $u_i = \frac{x - x_i}{h} = \frac{2.5 - x_i}{1}$

| $x_i$ | $u_i$ | $K(u_i)$          |
|-------|-------|-------------------|
| 1     | 1.5   | 0.1295            |
| 1.5   | 1     | 0.2419            |
| 2     | 0.5   | 0.3520            |
| 3     | -0.5  | 0.3520            |
| 3.5   | -1    | 0.2419            |
|       |       | $\Sigma = 1.3173$ |

$p(x) = \frac{1}{nh^n} \sum \frac{1}{\sqrt{2\pi}} e^{-\frac{(x-x_i)^2}{2h^2}}$   
 $= \frac{1}{5 \cdot 1} (1.3173)$   
 $= 0.2635$